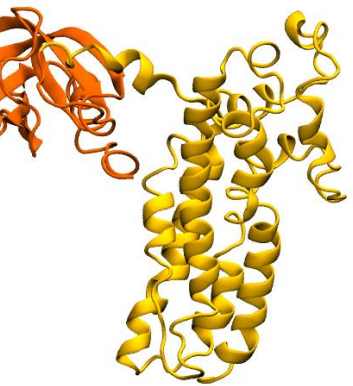
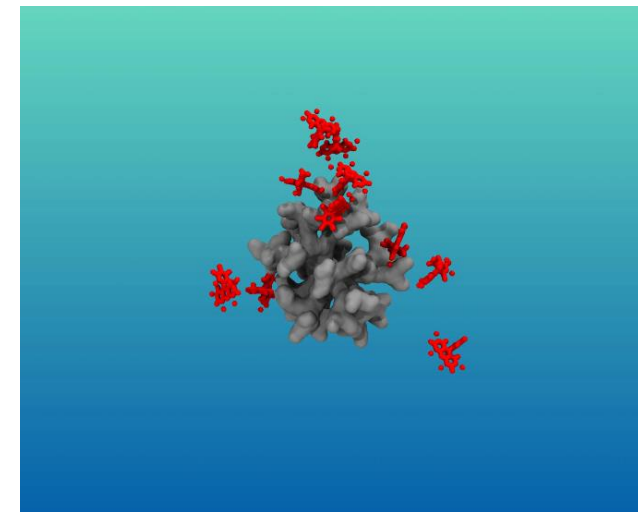
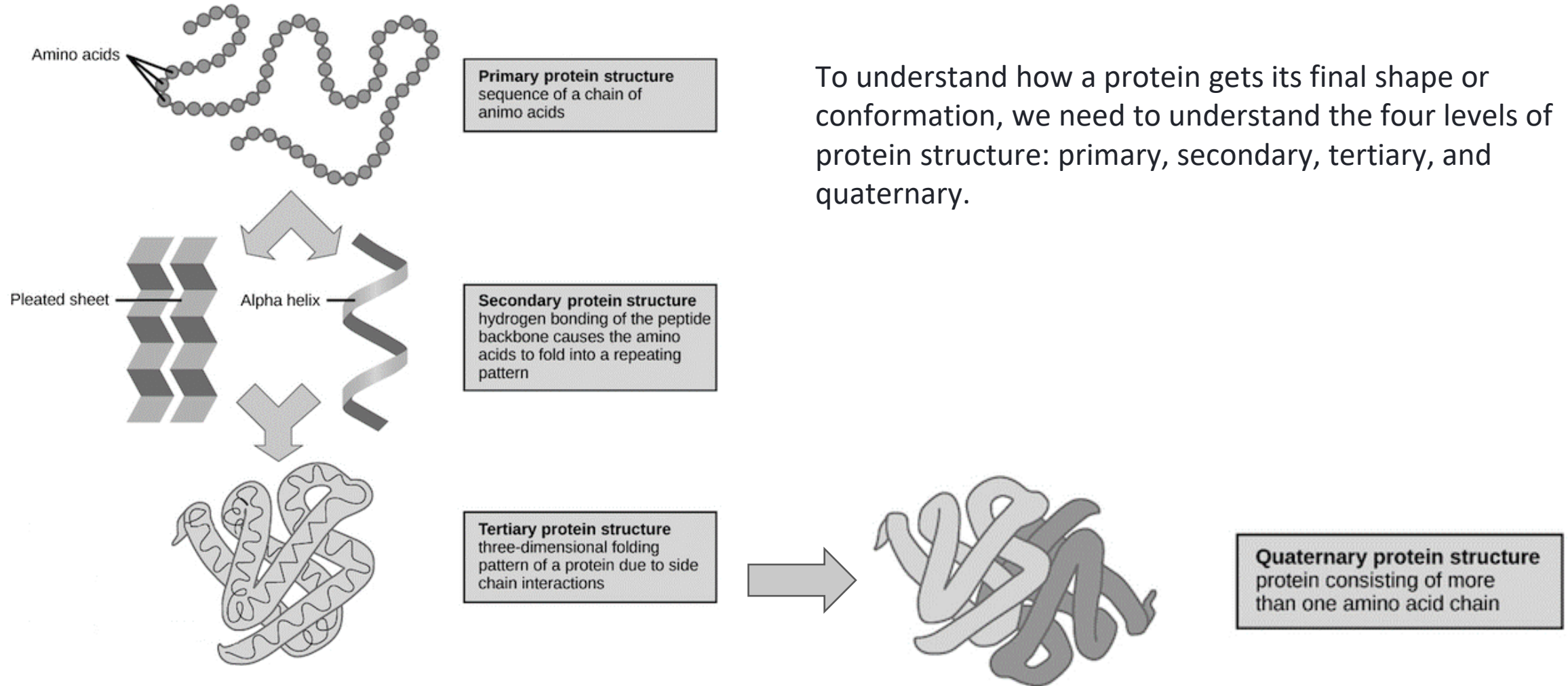


Introduction to molecular modelling



Order of Protein Structure



To understand how a protein gets its final shape or conformation, we need to understand the four levels of protein structure: primary, secondary, tertiary, and quaternary.

➤ **Protein databases:**

<https://www.rcsb.org/>



➤ **Ligand databases:**

<https://zinc.docking.org/>

<https://go.drugbank.com/>

<https://pubchem.ncbi.nlm.nih.gov/>

<https://www.ebi.ac.uk/chembl/>

<http://www.chemspider.com/>

ZINC15

PubC_hem

ChEMBL

DRUGBANK

ChemSpider
The free chemical database

Homology Modelling

Homology modeling build the molecular model of a **target** protein from its amino acid sequence and an experimental 3D structure of a related homologous protein, the **template**

Alignment

Query sequence Uniprot Q96Q42

MDKLKVIQQTFFEEISQSVLASLHEDFLWSMDD

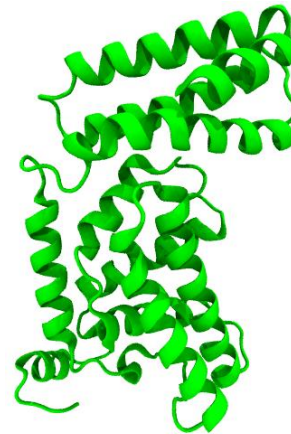
DIIEMDSKRVPRDKLACITKCSKHIFNAIK

MDKLKVIQQTFFEEISQSVLASLHEDFLWSMDD

...

/TPQMLCVPVNEIPEVSDMVVKAITDI

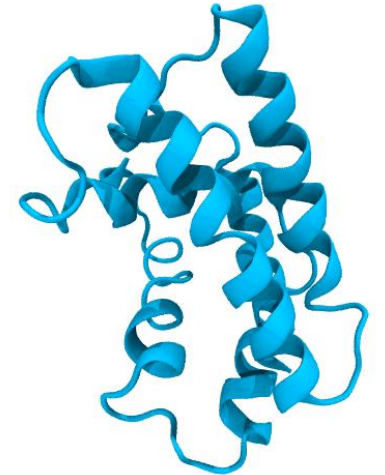
Template



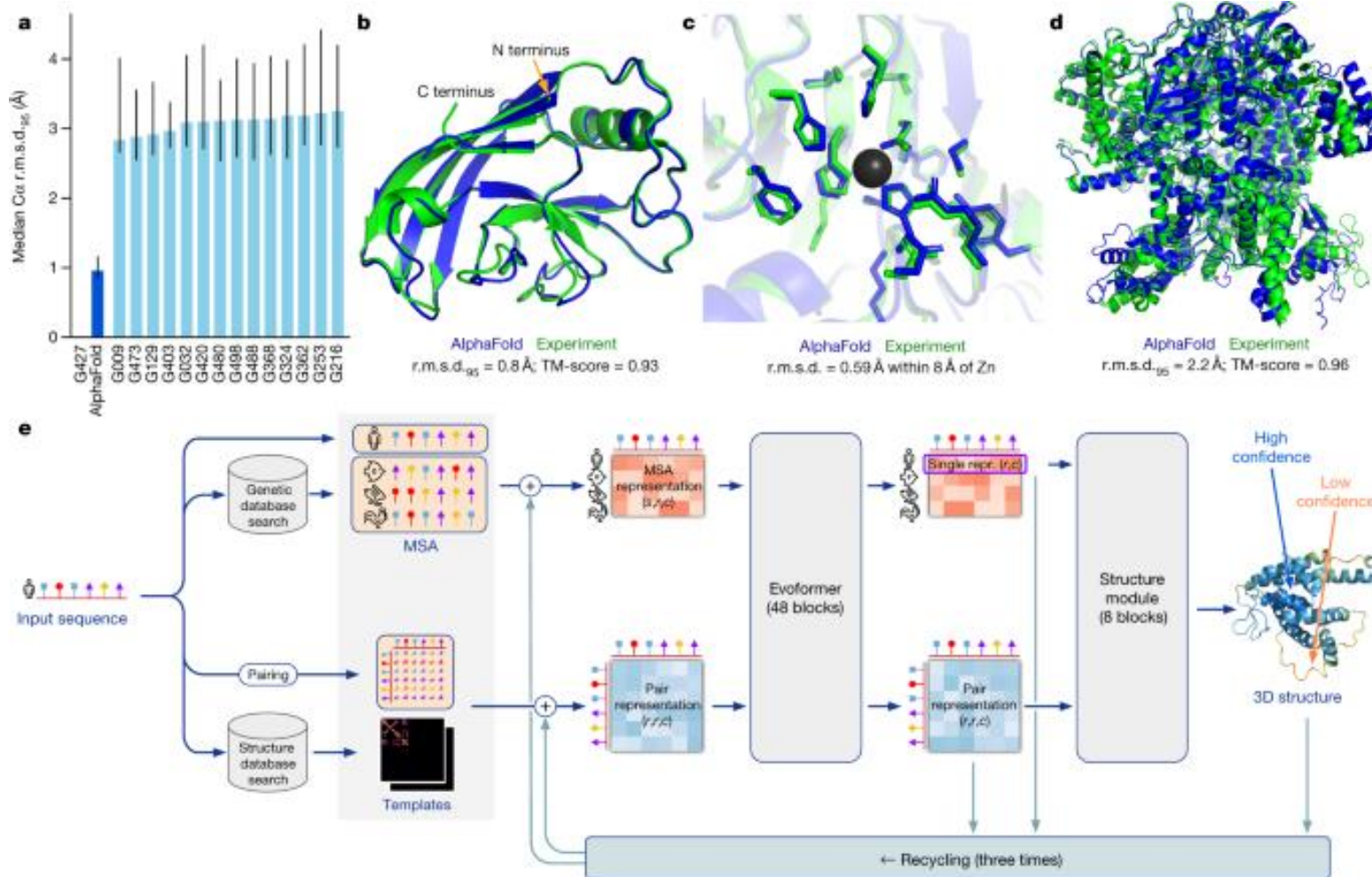
Modelling



Homology model



AlphaFold



[AlphaFold](https://alphafold.ebi.ac.uk/) is an AI system developed by DeepMind that predicts a protein's 3D structure from its amino acid sequence. It regularly achieves accuracy competitive with experiment.

DeepMind and EMBL's European Bioinformatics Institute ([EMBL-EBI](https://www.ebi.ac.uk/)) have partnered to create AlphaFold DB to make these predictions freely available to the scientific community.

<https://alphafold.ebi.ac.uk/>

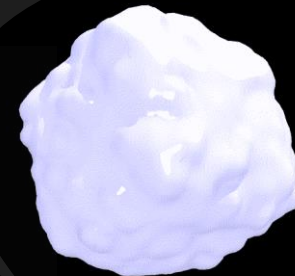
Play with alphafold2

[Link](#)

<https://www.nature.com/articles/s41586-021-03819-2>

MOLECULAR MODELLING AS A COMPUTATIONAL MICROSCOPE

$$\vec{F}_i = m a = m_i \frac{d^2 \vec{r}_i}{dt^2} = -\vec{\nabla} U(\vec{R})$$
$$U(\vec{R}) = \sum_{\text{bonds}} k_b^{\text{bond}} (r_i - r_o)^2 + \sum_{\text{angles}} k_\theta^{\text{angle}} (\theta_i - \theta_o)^2 + \sum_{\text{dihedrals}} k_\phi^{\text{dihedral}} (1 + \cos(n\phi_i + \delta_i)) + \sum_i \sum_{j \neq i} A \epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \sum_i \sum_{j \neq i} \frac{q_i q_j}{\epsilon r_{ij}}$$



Let's see an example



ACE2
+
Covid-19

Equation of motion

The equations that describe the temporal evolution of a physical system is called **equation of motion**. There are different equations of motions, which characterize the motion with different levels of approximation:

- Time-dependent Schrödinger's Equation for quantum-mechanical system
- **Newton's Equation for classical-mechanical system**
- Langevin's Equation for stochastic system

$$i \hbar \frac{\partial \psi}{\partial t}(r, t) = \hat{H} \psi(r, t)$$

$$m \frac{d^2 r}{dt^2} = -\lambda \frac{dr}{dt} + \eta(t)$$

Newton's Equation of motion

Molecules are quantum-mechanical systems whose motion should be described by Schrödinger's Equation. However, technical difficulties make solving Schrödinger's Equation for large systems impractical.

Therefore in classical molecular dynamics, the motion of a molecule is approximated by the laws of classical mechanics and by Newton's equation of motion. In its most simplistic form Newton's second law of motion states:

$$F_i = m_i a_i$$

where m_i is the mass of particle i , a_i is its acceleration.

But the force f_i can also be written as the derivative of the potential energy function V :

$$F_i = -\frac{\partial V}{\partial r_i}$$

where r_i is the position of particle

Potential energy function

Therefore, we need to define the potential of our molecular system ..

$$V(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_n) = \sum_{\text{bond}} \frac{1}{2} k_{b_n} (b_n - b_{0_n})^2 + \sum_{\text{angle}} \frac{1}{2} k_{\theta_n} (\theta_n - \theta_{0_n})^2 +$$

$$+ \sum_{\text{improper dihedral}} \frac{1}{2} k_{\xi_n} (\xi_n - \xi_{0_n})^2 + \sum_{\text{dihedral}} k_{\phi_n} [1 + \cos(m_n \phi_n - \delta_n)] +$$

$$+ \sum_{\text{nonbonded pairs}(ij)} \left(\left(\frac{C_{ij}^{(12)}}{r_{ij}^{12}} - \frac{C_{ij}^{(6)}}{r_{ij}^6} \right) + \frac{1}{4\pi\epsilon_0} \frac{q_i q_j}{\epsilon_r r_{ij}} \right)$$

non bonded interactions

$$U = \sum_{i < j} \sum 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right]$$

$$+ \sum_{i < j} \sum \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}}$$

$$+ \sum_{\text{bonds}} \frac{1}{2} k_b (r - r_0)^2$$

$$+ \sum_{\text{angles}} \frac{1}{2} k_a (\theta - \theta_0)^2$$

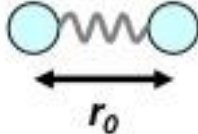
$$+ \sum_{\text{torsions}} k_\phi [1 + \cos(n\phi - \delta)]$$

bonded interactions

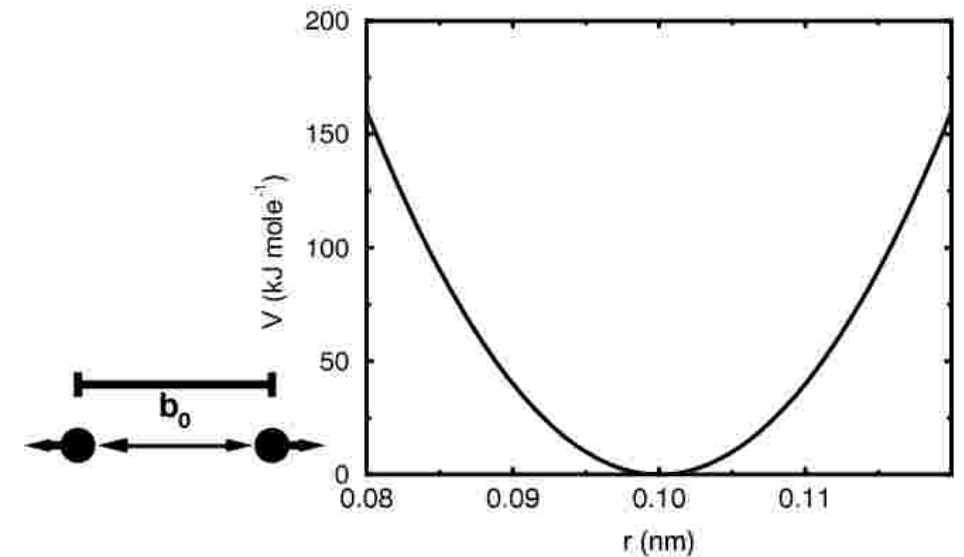
Bond Stretching Energy

$$\sum_{\text{bond}} \frac{1}{2} k_{b_n} (b_n - b_{0_n})^2$$

Unique k_b and b_0
assigned for each bond
pair, i.e. C-C, O-H

$$+ \sum_{\text{bonds}} \frac{1}{2} k_b (r - r_0)^2$$


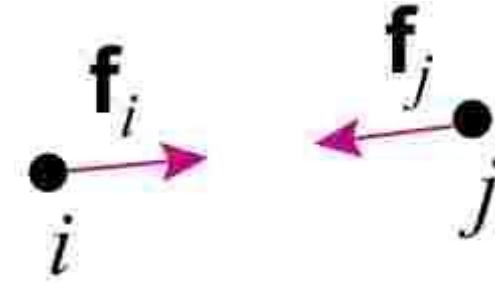
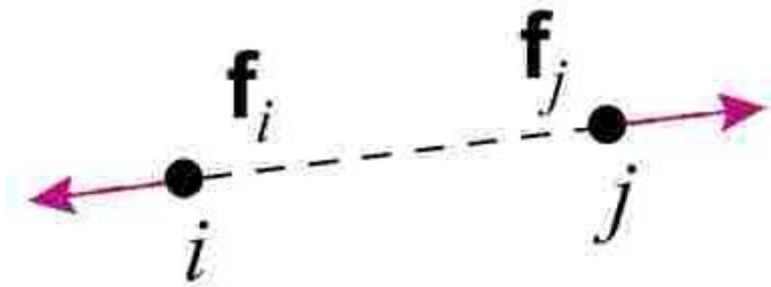
k_b is the spring constant of
the bond b_0 is the bond
length at equilibrium



Principle of bond stretching (left), and
the bond stretching potential (right).

Bond Stretching Force

If atom i and j are closer than b_0 , the bond force separates them

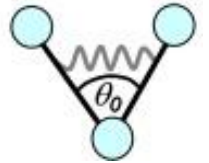


If atom i and j are farther than b_0 , the bond force draws them nearer

Bending Energy

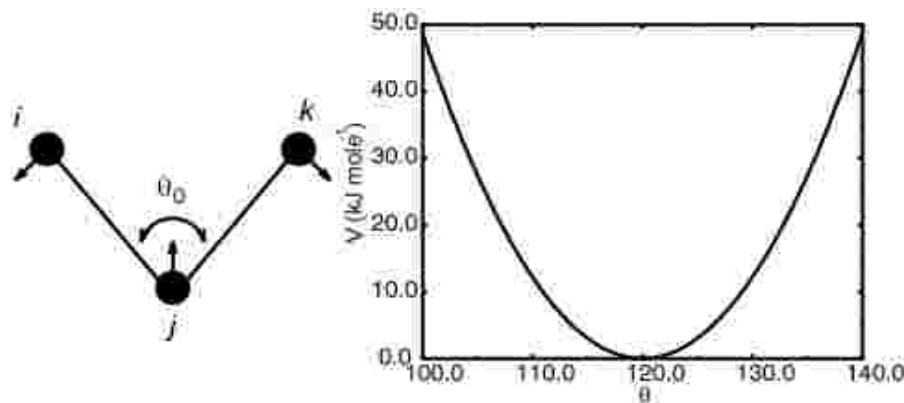
$$+ \sum_{\text{angle}} \frac{1}{2} k_{\theta_n} (\theta_n - \theta_{0_n})^2$$

$$+ \sum_{\text{angles}} \frac{1}{2} k_a (\theta - \theta_0)^2$$



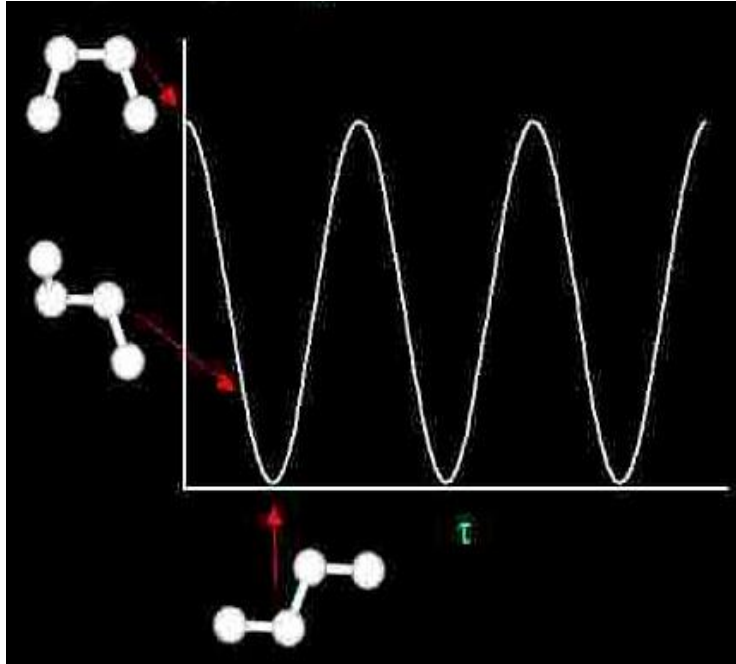
k_{θ} is the spring constant of the bending
 θ_0 is the angle bending at equilibrium

Unique parameters for
 angle bending are
 assigned to each
 bonded triplet of atoms
 based on their types
 (e.g. C-C-C, C-O-C, C-C-
 H, etc.)



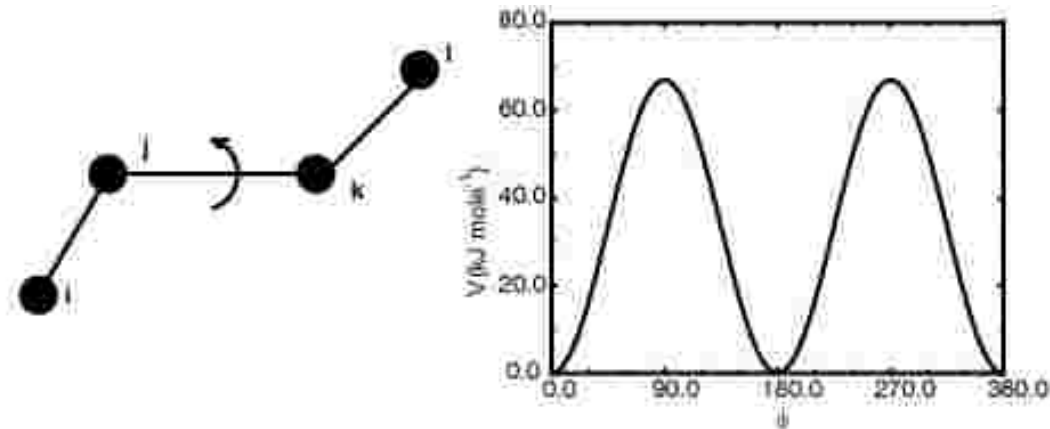
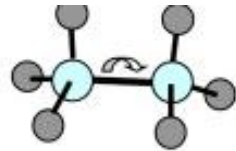
Principle of angle vibration (left) and the bond angle potential (right).

Torsional or Dihedral Energy



$$+ \sum_{\text{improper dihedral}} \frac{1}{2} k_{\xi_n} (\xi_n - \xi_{0n})^2 + \sum_{\text{dihedral}} k_{\phi_n} [1 + \cos(m_n \phi_n - \delta_n)]$$

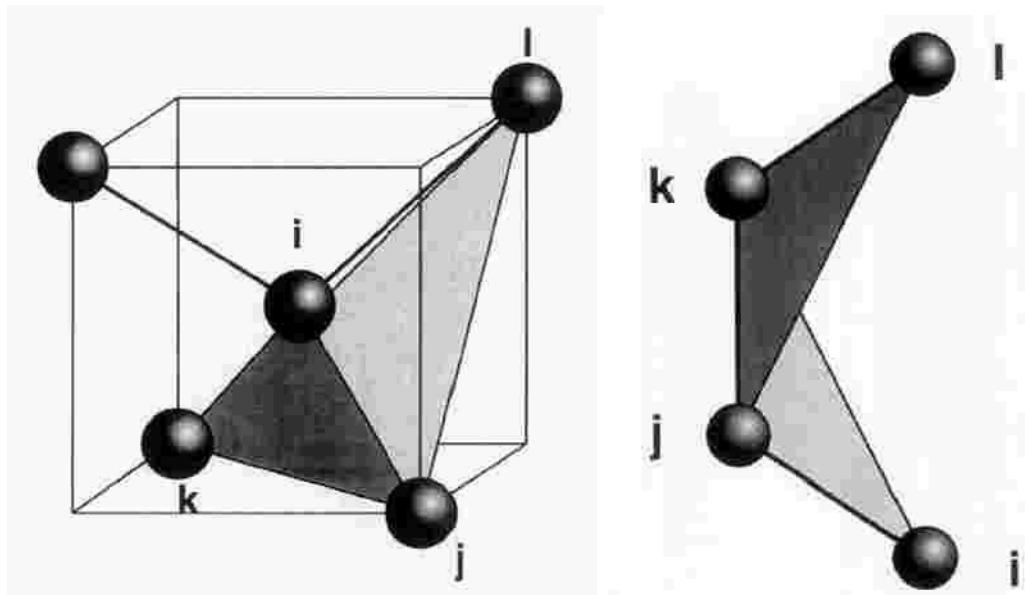
$$+ \sum_{\text{torsions}} k_{\phi} [1 + \cos(n\phi - \delta)]$$



Principle of proper dihedral angle (left, in *trans* form) and the dihedral angle potential (right).

Improper Dihedral Energy

The energy required to deform a group of atoms from its equilibrium angle, ξ_0 . Used for tetrahedral or planar groups

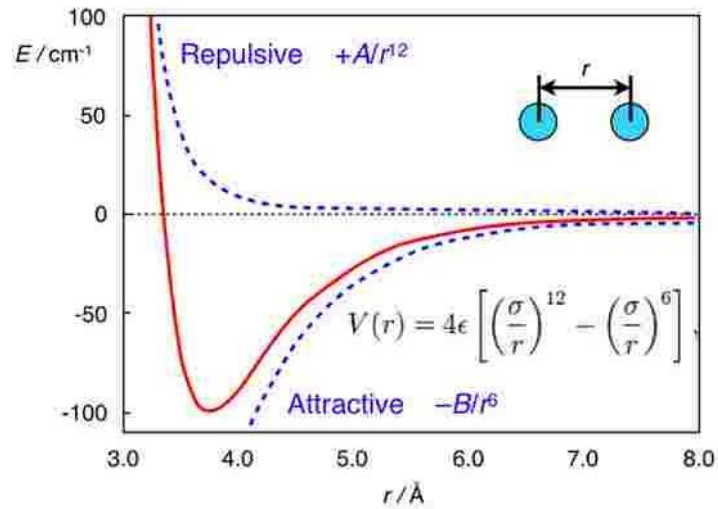


$$+ \sum_{\text{improper dihedral}} \frac{1}{2} k_{\xi_n} (\xi_n - \xi_{0_n})^2$$

Again this system can be modeled by a spring, and the energy is given by the Hookean potential with respect to the planar angle

Lennard Jones (Van der Waals) interactions

Sir John Lennard Jones

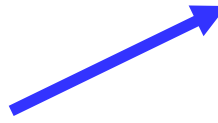


Johannes Diderik Van der Waals

LJ interactions

$$\sum_{\text{nonbonded pairs}(ij)} \left(\frac{C_{ij}^{(12)}}{r_{ij}^{12}} - \frac{C_{ij}^{(6)}}{r_{ij}^6} \right)$$

non bonded interactions



Electrostatic interactions

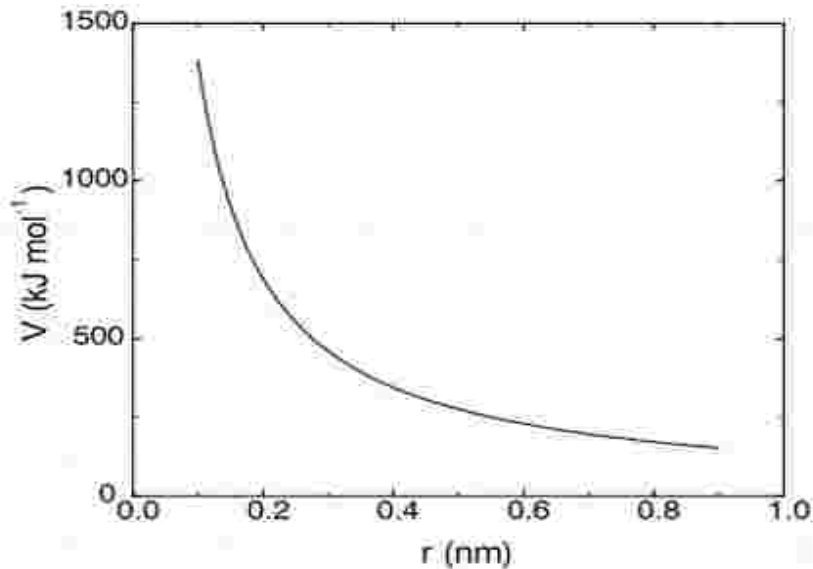
q_i and q_j are the partial atomic charges for atoms i and j , separated by a distance r_{ij} and ϵ_r is the relative dielectric constant:

Electrostatic

$$+ \frac{1}{4\pi\epsilon_0 \epsilon_r r_{ij}} \left(\frac{q_i q_j}{r_{ij}} \right)$$



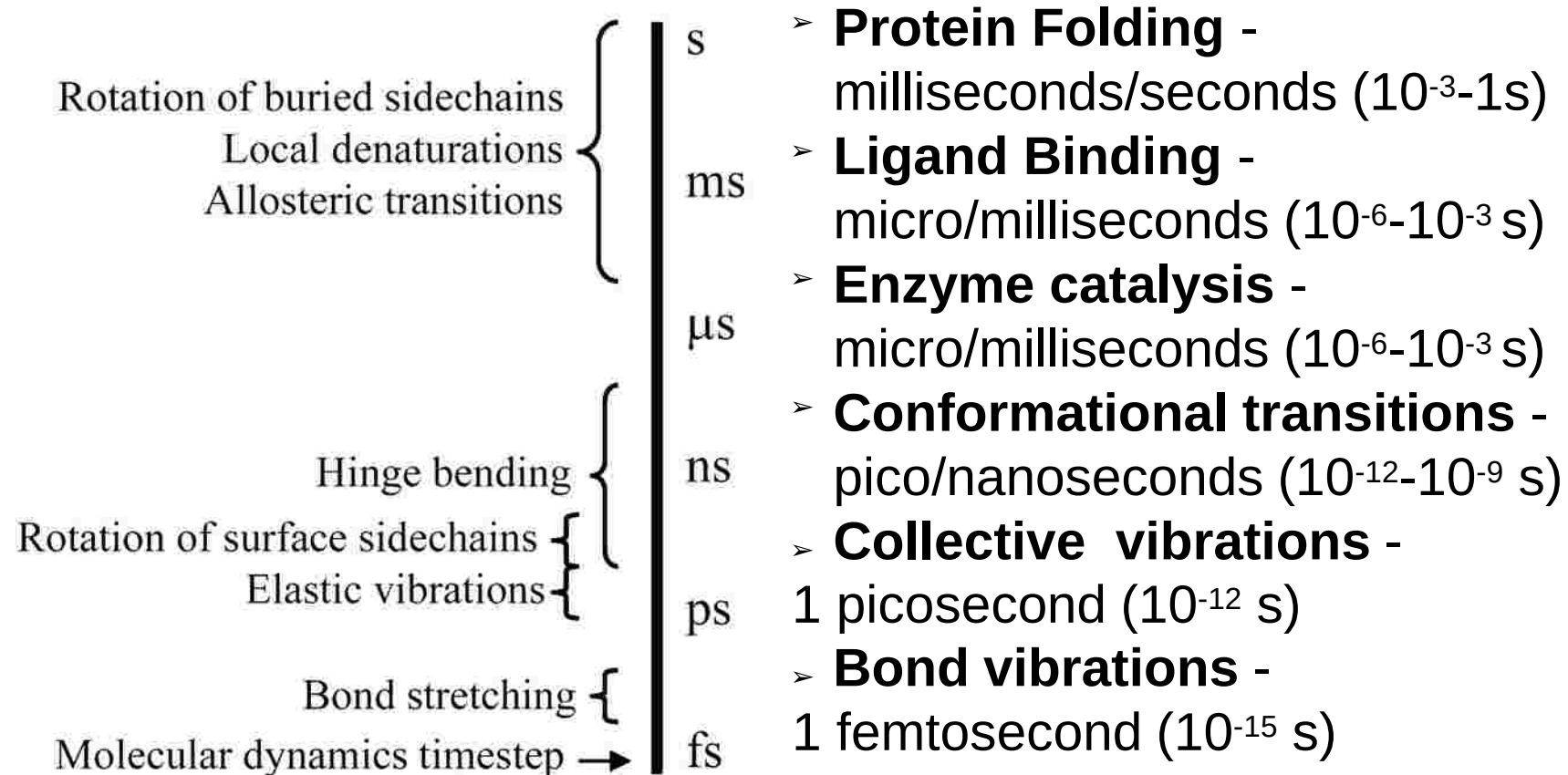
Charles Augustin de Coulomb



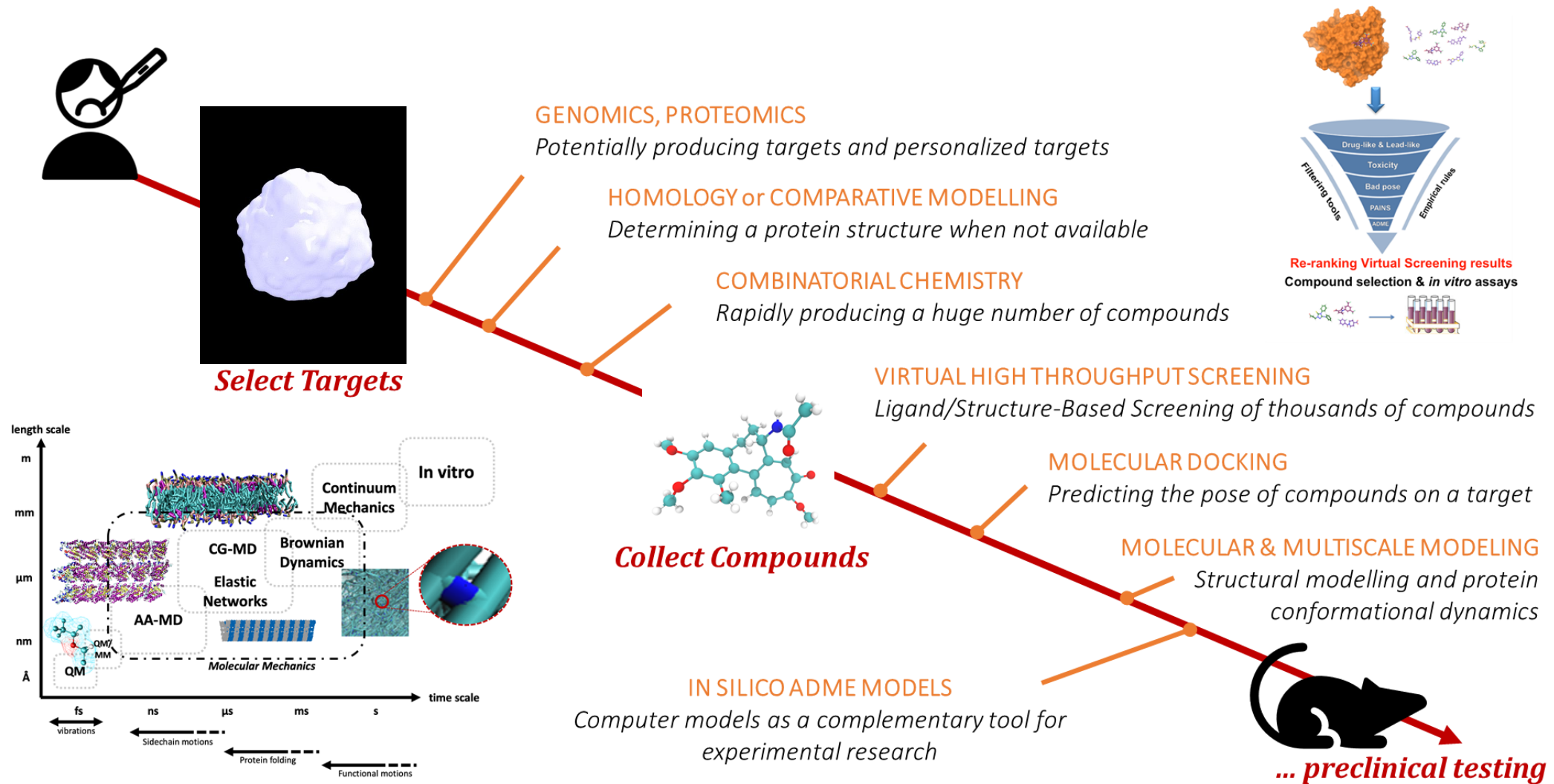
The Coulomb interaction (for particles with equal signed charge)

non bonded interactions

Timescale



The role of Molecular modelling

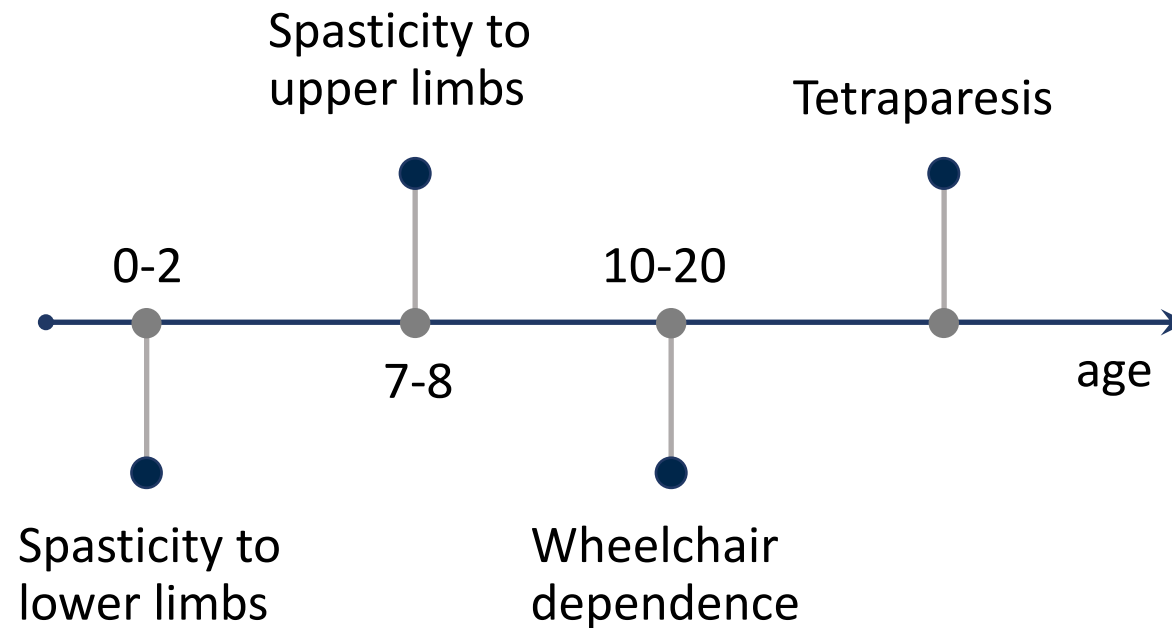


A real world application

IAHSP disease

IAHSP: Infantile onset Ascending hereditary spastic paraplegia

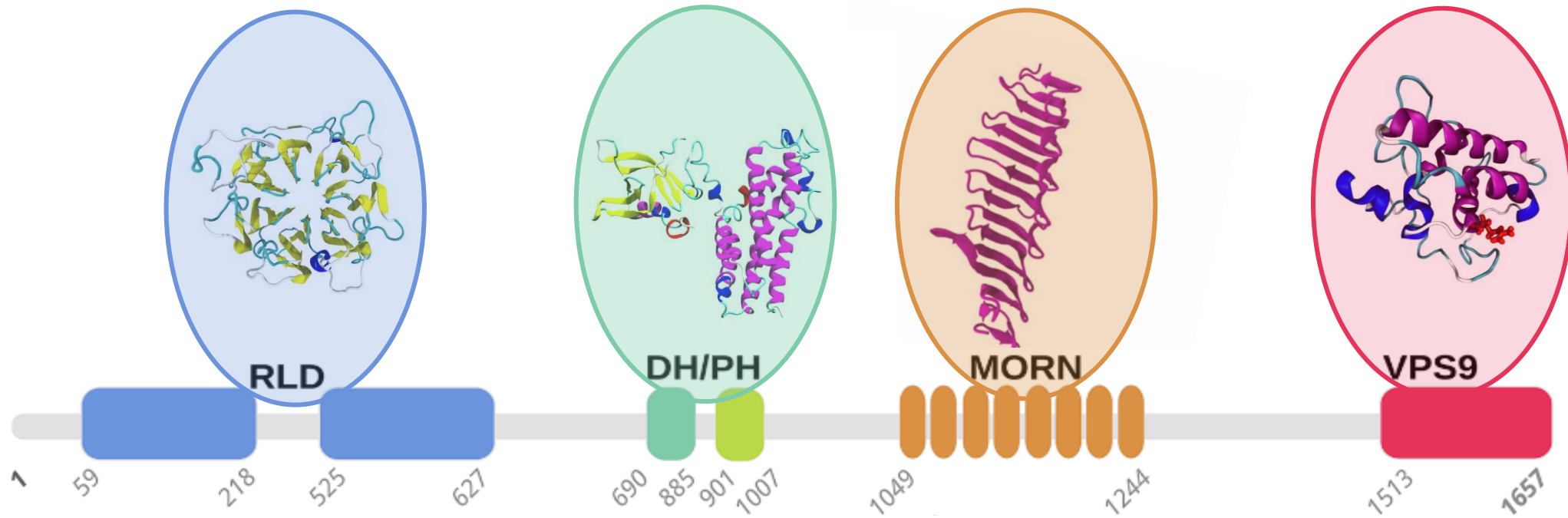
Rare neurodegenerative disease related to **mutation** on gene **ALS2**



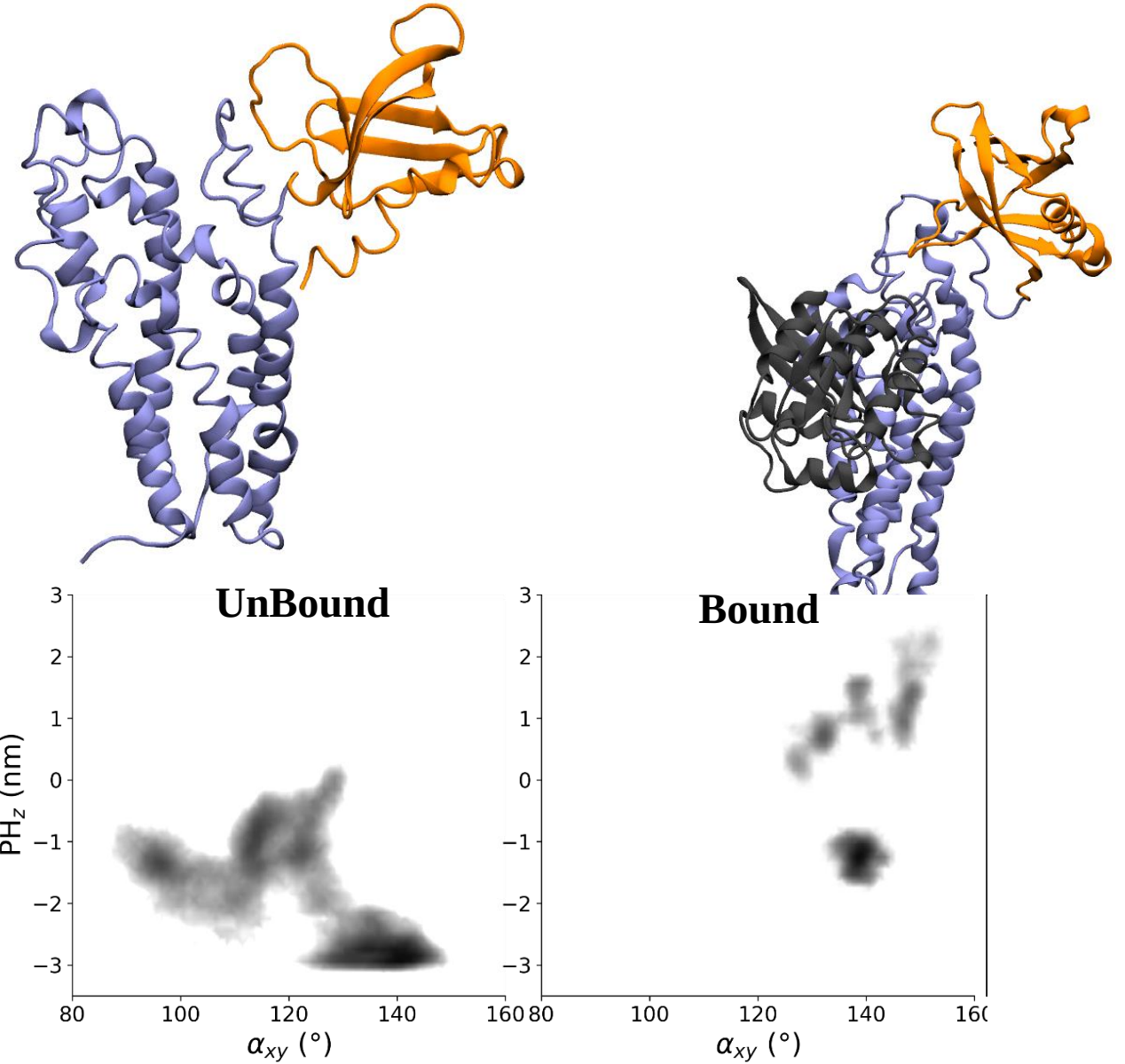


Hadano, S. *et al.* *Nat. Genet.* 29, 166–173 (2001)

Alsin homology models



DH/PH-rac1 interaction

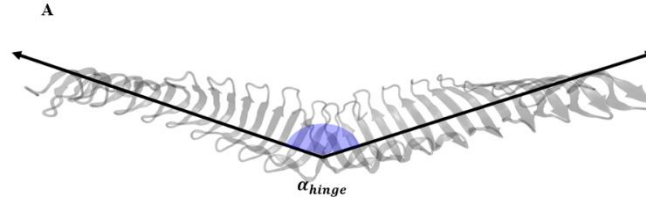


Dimerization of MORN Domain

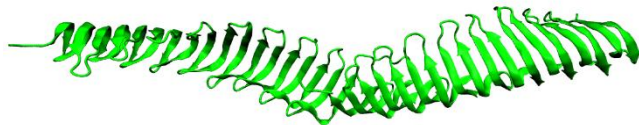
V-shaped conformation



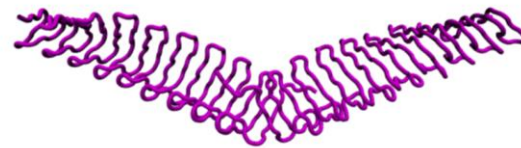
- **Linear** assembly → C-termini
- **V-shaped** assembly → C-termini + Zn ion



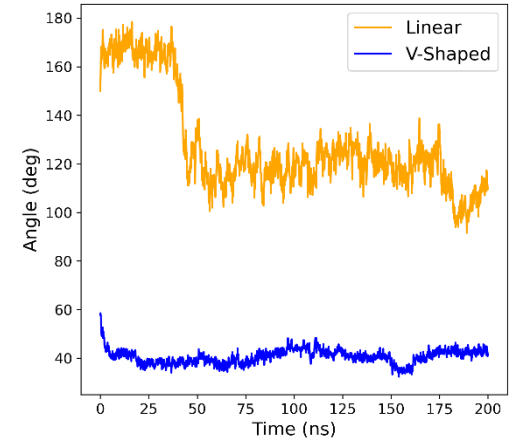
Linear conformation



TbMORN (6T4R)
TgMORN (6T6Q)



V-exp
N-ter ...
V-Alsin

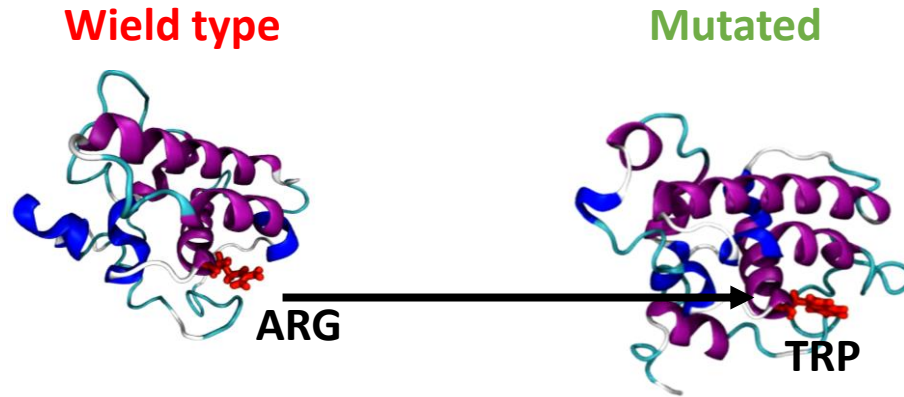


Zinc interaction domain



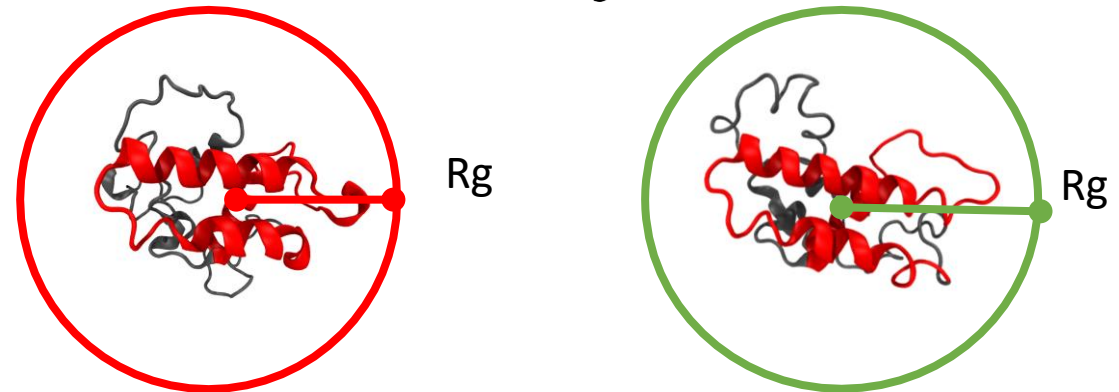
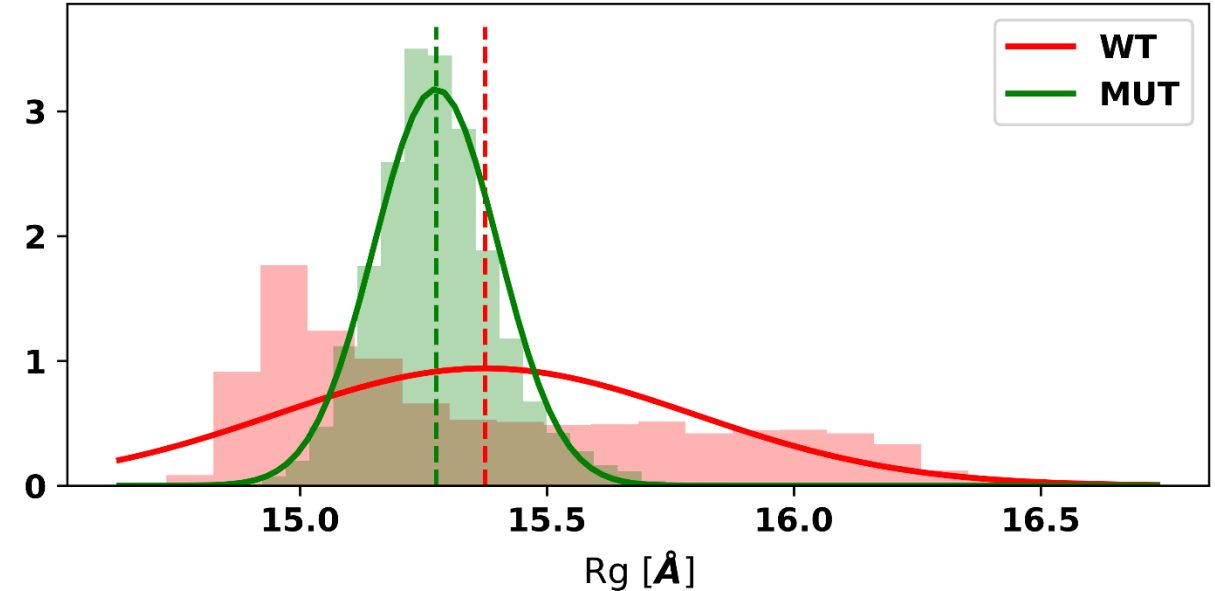
- ✓ **Asp 309** → **Asp 1217**
- ✓ **Glu 308** → **Glu 1216**
- × **Cys 307** → **Ser 1215**

VPS9 WT vs MUT



Radius of gyration (R_g) has been employed as a measure of the compactness of the structure.

$$R_g = \left(\frac{\sum_i ||r_i||^2 m_i}{\sum_i m_i} \right)^{\frac{1}{2}}$$





CRYSTAL

<https://crystal.m3b.it/>



November 2020

November 2021